Polynomial Division: The Box Method

Name: ______________________________ **Date:** __________ **Period:** _____

The Connection: Division is the **inverse** of multiplication!

Multiplication (Forward)	Division (Backward)
Fill in the box → Add diagonals → Product	Know the product → Work backwards → Find missing factor

In division: **Dividend** ÷ **Divisor** = **Quotient**. The divisor goes on the **left**, quotient goes on **top**.

Example 1: $(x^3 + 5x^2 + 8x + 4) \div (x + 1)$

$\times$	x^2	$4x$	4
x	x^3	$4x^2$	$4x$
$+1$	x^2	$4x$	4

Add diagonals: $x^3 + 5x^2 + 8x + 4$ ✓

Step 1: Set up the box. Divisor on left.
Step 2: First term: $x^3 \div x = x^2$ (top)
Step 3: Fill column: $x^2 \cdot 1 = x^2$
Step 4: Next: $(5x^2 - x^2) \div x = 4x$
Step 5: Continue until complete.
Step 6: Check—does it rebuild the dividend?

Quotient: $\boxed{x^2 + 4x + 4}$ **Remainder:** $\boxed{0}$

Example 2: $(2x^4 + 3x^2 - 10) \div (x + 2)$ ← *Missing x^3 and x terms!*

$\times$	$2x^3$	$-4x^2$	$11x$	-22
x	$2x^4$	$-4x^3$	$11x^2$	$-22x$
$+2$	$4x^3$	$-8x^2$	$22x$	-44

Diagonals: $2x^4 + 0x^3 + 3x^2 + 0x - 44$
But we need -10, not -44!

Key step: Rewrite with placeholders:
$(2x^4 + \mathbf{0x^3} + 3x^2 + \mathbf{0x} - 10) \div (x + 2)$
Step 1: $2x^4 \div x = 2x^3$
Step 2: $2x^3 \cdot 2 = 4x^3$; need $0x^3$, so next: $(0 - 4x^3) \div x = -4x^2$
Step 3: Continue pattern...
Step 4: Last cell doesn't match? That's the **remainder**!

Quotient: $\boxed{2x^3 - 4x^2 + 11x - 22}$ **Remainder:** $-10 - (-44) = \boxed{34}$

You Try: $(3x^3 + 10x^2 + x - 6) \div (x + 2)$

$\times$			
x			
$+2$			

Quotient: ____________________

Remainder: ________________